
Show that the probability per unit time λ given by¹

$$\lambda = \omega(\alpha) \frac{v}{2R} e^{-G} \quad (1)$$

indeed leads to the Geiger-Nuttall relation $\log_{10} t_{1/2} = \alpha + bZE_{\alpha}^{-1/2}$, where the constant α depends on Z but b does not.

Proof. Starting from

$$\lambda = \omega(\alpha) \frac{v}{2R} e^{-G}, \quad t_{1/2} = \frac{\ln 2}{\lambda},$$

we have

$$\ln t_{1/2} = \ln \left(\frac{2R \ln 2}{\omega(\alpha)v} \right) + G.$$

Since $G \propto ZE_{\alpha}^{1/2}$, then

$$G = \kappa \frac{Z}{\sqrt{E_{\alpha}}},$$

where κ is some constant. Therefore

$$\log_{10} t_{1/2} = \log_{10} \left(\frac{2R \ln 2}{\omega(\alpha)v} \right) + \frac{\kappa}{\ln 10} \frac{Z}{\sqrt{E_{\alpha}}}.$$

Defining

$$\alpha = \log_{10} \left(\frac{2R \ln 2}{\omega(\alpha)v} \right), \quad b = \frac{\kappa}{\ln 10},$$

gives

$$\boxed{\log_{10} t_{1/2} = \alpha + bZE_{\alpha}^{-1/2}} \quad (2)$$

Since $\omega(\alpha) = \omega(\alpha, Z)$, $\alpha = \alpha(Z)$, while b is independent of Z . $\square$

¹ $\omega(a)$ depends on Z